

01_Assignment_Yuliia_Chernysheva

August 28, 2026

1 CIS 2267 - Assignment 1 - Review

Complete the following tasks using the Python programming language.

1.1 1.

Write a function to calculate the minimum, maximum, total, and average values of a list. Return these four values as a tuple.

Display a menu like this:

1. Enter a value
2. Display statistics
3. Exit

Get user input and take the appropriate action. Repeat until menu choice 3 is chosen.

Display statistics to one decimal place. Beware of division by zero.

```
[1]: def display_menu():
    print("1. Enter a value")
    print("2. Display statistics")
    print("3. Exit")

def stats(list1):
    if len(list1) == 0:
        return (0,0,0,0)

    min_value = list1[0]
    max_value = list1[0]
    total = 0

    for i in list1:
        total += i
        if i < min_value:
            min_value = i
        if i > max_value:
            max_value = i

    avg = total / len(list1)
```

```

    return (min_value, max_value, total, avg)

list_values = []
choice = ""

while choice != "3":
    display_menu()
    choice = input("Choose an option: ")
    if (choice == "1"):
        input_value = input("Enter a value => ")
        list_values.append(float(input_value))
    elif (choice == "2"):
        print("-"*25, "STATISTICS", "-"*25)
        mi, ma, t, a = stats(list_values)
        print("\n=> Minimum = {:.1f}; Maximum = {:.1f}; Total = {:.1f}; Average_
=> {:.1f}\n".format(mi, ma, t, a))
        print("-"*62)

```

1. Enter a value
2. Display statistics
3. Exit

Choose an option: 45

1. Enter a value
2. Display statistics
3. Exit

Choose an option: yy

1. Enter a value
2. Display statistics
3. Exit

Choose an option: 2

----- STATISTICS -----

=> Minimum = 0.0; Maximum = 0.0; Total = 0.0; Average = 0.0

-
1. Enter a value
 2. Display statistics
 3. Exit

Choose an option: 1

Enter a value => 4

1. Enter a value
2. Display statistics
3. Exit

```
Choose an option: 1
Enter a value => 5
```

1. Enter a value
2. Display statistics
3. Exit

```
Choose an option: 1
Enter a value => 1
```

1. Enter a value
2. Display statistics
3. Exit

```
Choose an option: 2
```

```
----- STATISTICS -----
```

```
=> Minimum = 1.0; Maximum = 5.0; Total = 10.0; Average = 3.3
```

- ```

```
1. Enter a value
  2. Display statistics
  3. Exit

```
Choose an option: 3
```

## 1.2 2.

Using **Markdown** syntax, write a short summary of the data analysis process. Use headings, bulleted lists, and bold formatting. Include a hyperlink to a web page with further information.

Before starting any analysis, two planning steps are essential:

- **Set your goals:** Decide whether you want to answer specific questions or learn general information.
- **Define your target audience:** Determine who will receive your findings.

Once goals and audience are defined, a typical project proceeds through **five phases**, which often overlap in practice:

## 2 Get the data

- Find the required data on a website, database, or spreadsheet.
- Read or import it into a structured format like a Pandas DataFrame.

## 3 Clean the data

- Remove unnecessary rows or columns.
- Handle missing or invalid values.
- Correct data types.

*Skipping this phase can compromise the accuracy of your results.*

## 4 Prepare the data

- Shape the data for analysis.
- Add calculated, derived, or aggregated columns.
- Make preliminary visualizations to understand the data structure.

## 5 Analyze the data

- Group and aggregate the data to uncover new perspectives.
- Build visualizations to find relationships and insights.
- Create predictive models.

## 6 Visualize the data

Polish and enhance final visualizations so that they communicate findings clearly and quickly, even to non-technical audiences.

### 6.1 Read

[Using pandas and Python to Explore Your Dataset.](#)

[An introduction to seaborn](#)

### 6.2 Learn more

[Markdown Help](#)

[ ]: